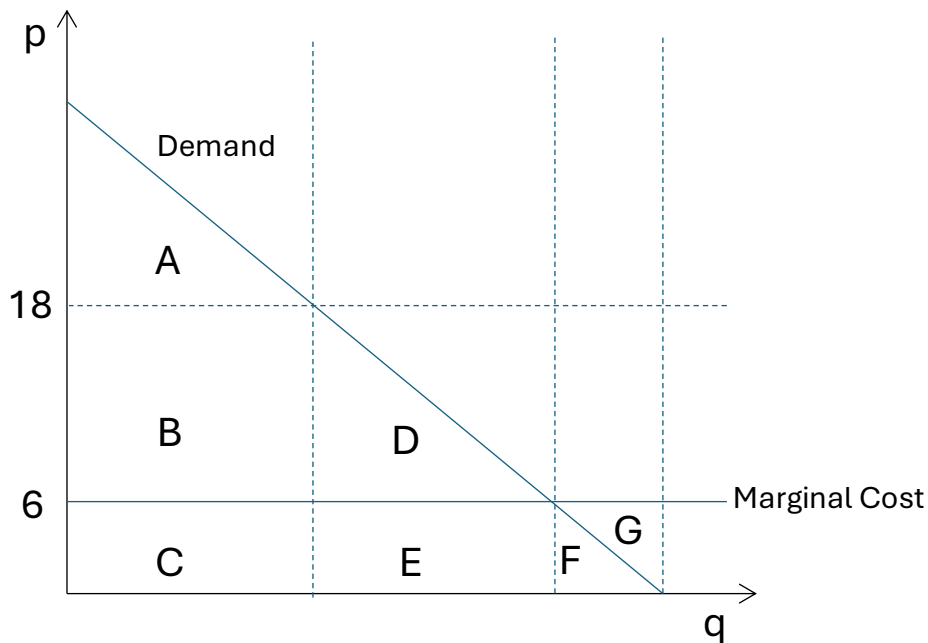


4 *Crossfit*. Crossfit is a popular group workout. All of Crossfit's 1000 target customers in Philadelphia are identical and have a monthly demand given by:

$$p = 30 - q$$

The marginal cost of an additional person in the class is \$6. The demand and marginal cost is depicted on the picture below.



(You are not required to show supporting work for this question.)

(a) (1 point). Cross-fit determines that their best uniform, per-class price is \$18 per class.

- i. How many times would the customer go? $q^* = 30 - 18 = 12$
- ii. Total amount each customer pays
(list the letters from the picture above): $B + C$
- iii. Company profit, per customer (list the letters): B
- iv. Consumer surplus (list the letters): A
- v. Deadweight loss (list the letters): D

(b) (1 point). Cross-fit decides to price instead with a two-part tariff. Indicate the area on the graph that would correspond to the membership fee and to the usage fee each customer would pay in an optimal two-part tariff, then describe profit and usage:

i. Area for membership fee (list the letters): A + B + D

ii. Area for usage fee (list the letters): C + E

iii. Company profit (list the letters): A + B + D

iv. How many times would the customer go? 30 - q = 6, q* = 24